

Governance as Inference-Time Capability Substitution: Reducing Tokens per Correct Software-Engineering Task with Deterministic Requirements and Verification

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<https://github.com/layer1labs/specsmith>

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Abstract

Language-model comparisons usually hold the agent scaffold fixed and vary the model. We study the complementary systems question: can deterministic governance compensate for measured model capability in repository-level software tasks? We introduce GovernanceBench, an executable benchmark and matched 2×2 design in which a lower-tier and frontier model each run with an ungoverned agent loop and with SPECSMITH FULL governance. The controller exposes a stable five-tool interface, retrieves only requirement-linked context, enforces visible tests without exposing hidden acceptance oracles, and audits every trace. Our primary efficiency measure is tokens per correct answer; correctness is a hard non-inferiority constraint. In the preregistered eight-task, ten-repetition experiment, GPT-5.6 Terra with FULL governance passed 80/80 cells at 11,737 tokens per correct answer, while GPT-5.6 Sol in the raw loop passed 65/80 at 28,023. The TPCA ratio was 0.419 (fixed-suite 95% interval 0.358–0.485; task-cluster interval 0.212–0.708). Both preregistered mixed-suite substitution gates passed. In a prespecified coding-only sensitivity, Terra FULL passed 60/60 at 15,649 TPCA versus Sol raw 55/60 at 30,250, but its correctness lower bound was -5.7 percentage points, just outside the -5 -point non-inferiority margin. Published 8B and mini-model routes failed admission. The evidence supports task-conditional substitution for the exact API routes and benchmark distribution, not universal model equivalence or a claim that small models replace frontier models.

1 Introduction

Repository-level coding agents must localize requirements, edit multiple files, run tools, interpret failures, and decide when evidence is sufficient. Benchmarks such as SWE-bench expose this interaction complexity [4], while SWE-agent shows that the agent-computer interface itself can materially change performance [12]. Consequently, “which model is best?” is incomplete: practical outcomes depend on the joint system of model, context policy, tools, validators, and stopping rules.

This paper asks whether a governance controller can act as *inference-time capability substitution*. The target is not to make a lower-tier route intrinsically equivalent to a frontier model. It is to determine whether a lower-tier model embedded in a deterministic, evidence-seeking controller can preserve task correctness while consuming fewer tokens or dollars than a frontier model in an ungoverned loop.

The question connects three literatures. Interactive reasoning methods ground generation in external actions and feedback [6, 8, 13]. Simpler software-agent pipelines can outperform more elaborate loops [11]. Model cascades and routers exploit heterogeneous cost–quality tradeoffs [2, 7]. SPECSMITH differs by routing not only between models but between probabilistic generation and deterministic governance operations: requirement lookup, scoped retrieval, test execution, safety refusal, evidence compression, and stopping.

We make four contributions:

1. a matched 2×2 experimental design separating model capability from governance lift;
2. a primary metric, tokens per correct answer, coupled to a preregistered correctness non-inferiority gate;
3. deterministic fixed-suite and task-cluster uncertainty analysis plus machine-readable claim gates; and
4. a versioned compact evidence artifact containing per-cell outcomes, workflow provenance, negative results, and a post-run weakness audit, backed by versioned tasks, validators, and hidden acceptance oracles.

2 Related Work

Executable software-engineering benchmarks. SWE-bench evaluates issue resolution against executable repository tests [4]. Static benchmark contamination and memorization are material threats [5]; SWE-bench-Live addresses freshness and repository diversity with continuously updated tasks and containerized environments [14]. GovernanceBench is smaller and synthetic by design. It

cannot estimate population-level coding ability, but its fixtures make controller ablation, exact counterfactual matching, and trace-level token accounting practical.

Agent interfaces and feedback. SWE-agent attributes substantial gains to a model-oriented interface [12]. ReAct interleaves reasoning with environmental actions [13]; Reflexion and Self-Refine use iterative feedback without parameter updates [6, 8]. Agentless demonstrates that constrained localization, repair, and validation can be competitive at low cost [11]. These findings motivate a compact tool surface and controller-owned validation rather than a large catalogue of agent skills.

Cost-aware model selection. FrugalGPT and RouteLLM demonstrate that heterogeneous models can be cascaded or routed to improve cost-quality tradeoffs [2, 7]. Our unit of analysis is a complete software task, and our router signal is not a learned preference score. Admission and post-run audit use executable correctness, requirements, scope boundaries, token receipts, and task-specific historical envelopes.

Reproducibility and variance. Benchmark point estimates can be dominated by uncontrolled variation [1]. Recent analysis of 640 LLM-for-software-engineering papers reports persistent artifact, environment, and versioning gaps [9]. We therefore repeat every release cell, pin task and controller revisions to one commit, reject missing provider cells, record model routes and tool-schema hashes, and publish interval estimates rather than relying on means alone.

3 System

3.1 Governance controller

SPECSMITH FULL wraps a language model with six mechanisms:

1. **Requirement contracts:** accepted requirements, linked tests, scope, risk, and epistemic constraints are loaded before edits.
2. **Bounded context:** versioned requirement-linked files may be preloaded; all other content is retrieved just in time.
3. **Stable interface:** every model turn sees the same five tools: batched and scalar read/write operations plus completion.
4. **Deterministic gates:** ambiguous or destructive tasks can stop without an LLM call; coding completion requires current public validation evidence.
5. **Evidence compression:** unchanged reads become digest receipts, superseded observations are removed,

and only active milestone state remains in long-horizon context.

6. **Hidden scoring:** acceptance oracles run only after the model stops and never supply repair hints.

The design intentionally excludes broad repository preloads, mutable tool schemas, evaluator leakage, and unbounded self-reflection. A deterministic post-run audit identifies correctness, scope, retrieval, repair, tool-schema, and cost weaknesses and selects the next admissible experiment.

3.2 Promotion funnel

New model routes receive one long-horizon governed admission. Controller changes receive one cell on each affected path. Successful candidates advance to a four-task, five-repetition 2×2 screen. Only then may the complete eight-task, ten-repetition substitution profile run. This funnel prevents a failed candidate from becoming persuasive merely through additional spend.

4 GovernanceBench

4.1 Tasks and conditions

The release grid contains eight tasks: two feature/bug tasks, two deterministic clarification or safety gates, three cross-file feature/refactor tasks, and one long-horizon polyglot incident console spanning Python, Go, TypeScript/React, JSON Schema, browser tests, and documentation. Six tasks require code changes; two assess whether the system declines or clarifies unsafe or underspecified work.

The substitution experiment crosses two models with two conditions:

Cell	Purpose
Lower-tier raw	lower-tier capability baseline
Lower-tier FULL	candidate governed system
Frontier raw	substitution target
Frontier FULL	frontier governance control

All four cells use identical task fixtures, repetition identifiers, provider APIs, temperature policy, turn cap, completion cap, and hidden oracles. Provider errors invalidate a cell rather than counting as free failures.

4.2 Metrics

For runs $i = 1, \dots, n$, with token count t_i , cost c_i , and correctness $y_i \in \{0, 1\}$, we report

$$\text{TPCA} = \frac{\sum_i t_i}{\sum_i y_i}, \quad \text{CoP} = \frac{\sum_i c_i}{\sum_i y_i}. \quad (1)$$

This formulation charges failed attempts to the expected cost of obtaining a correct answer. Raw tokens are primary because provider prices and cache discounts change; estimated dollars are secondary.

Correctness is evaluated by deterministic lint, public tests, task-specific contract validators, and an evaluator-isolated acceptance oracle. We also record turns, wall time, files read and written, scope expansion, first-pass rate, provider-reported cache tokens, and a stable tool-schema hash.

4.3 Hypotheses and claim gates

The release profile and thresholds were committed at revision 75a8c79 before workflow dispatch. The primary comparison is lower-tier FULL minus frontier raw:

H1: correctness non-inferiority. The lower 95% bound on the pass-rate difference is at least -5 percentage points.

H2: token superiority. The upper 95% bound on the TPCA ratio is below 1.

H3: cross-task robustness. H1 and H2 also hold when task identifiers, not only repetitions, are resampled.

H4: governance lift. Lower-tier FULL improves correctness or TPCA relative to the same lower-tier raw model.

We use Wilson score bounds for binomial rates [10] and a deterministically seeded, 10,000-sample hierarchical bootstrap [3]. The fixed-suite analysis resamples repetitions within each task. The task-cluster analysis additionally resamples task IDs; it tests robustness to resampling the benchmark tasks but cannot establish performance on fresh repositories. A result can therefore support fixed-suite substitution while failing cross-task substitution. No result supports universal model equivalence.

5 Results

Preregistered release inference. The lower-tier FULL/frontier-raw TPCA point ratio was 0.419. Its fixed-suite 95% interval was 0.358–0.485; the task-cluster interval was 0.212–0.708. The corresponding correctness-difference intervals were 3.3–34.1 pp and -4.4 – 54.7 pp. The machine-readable release gate passed. Fixed-suite substitution was supported, and cross-task substitution was supported.

Coding-only sensitivity. After excluding deterministic clarification and safety gates, lower-tier FULL passed 60/60 coding cells at 15,649 TPCA; frontier raw passed 55/60 at 30,250 TPCA. The TPCA ratio was 0.517, with fixed-suite 95% interval 0.437–0.608 and task-cluster interval 0.328–0.870. The correctness-difference intervals were

-5.7 – 26.1 pp and -6.0 – 31.8 pp. Coding-only fixed-suite substitution was not supported; this sensitivity analysis was specified for interpretation after the mixed-suite release gate and is not relabelled as the primary endpoint.

The primary mixed-suite point estimate corresponds to 58.1% lower TPCA and 60.1% lower estimated cost per correct answer for Terra FULL than Sol raw. The task-cluster correctness lower bound was -4.4 percentage points, clearing the preregistered -5 -point margin by 0.6 points. The coding-only sensitivity went in the same observed direction but did not clear that margin; deterministic clarification and safety gates therefore matter to the strength of the primary claim. We report both endpoints rather than relabelling the sensitivity after observing the result.

The four-task screen demonstrates why both uncertainty views are needed. Its point TPCA ratio is 0.653. The fixed-suite 95% interval is 0.539–0.784, while the task-cluster interval is 0.339–1.396. Terra FULL is less token-efficient than Sol raw on T1 and T10 but substantially more efficient on T13 and T28. The aggregate is therefore useful routing evidence, not a universal replacement result.

6 Trace-Guided Ablations

The benchmark audit converts failures and expensive correct runs into bounded controller experiments. Three examples illustrate the loop:

- A T10 controller denied reads of an imported model and dependency manifest. Adding exactly those files reduced one admission from 63,128 tokens and 11 turns to 9,305 tokens and two turns.
- Repeated T11 first actions justified a bounded preload, reducing a diagnostic from 23,472 tokens and six turns to 11,210 tokens and two turns.
- Applying the same idea to T13 increased tokens from a 14,424-token broad-run mean to 16,190 in admission. The preload was removed.

The negative T13 result is important: more context is not monotonically beneficial, and governance optimization itself requires admission controls.

Open-model admissions provide a second negative control. Qwen3.6-35B-A3B completed the long-horizon task with a stable tool schema but used 67,701 tokens, 3.87 times the governed frontier anchor. It was not promoted to repeated spend. Correctness alone is insufficient for an efficiency claim.

A published-size Llama 3.1 8B route failed three governed T28 admissions. Strict recovery of its text-serialized function calls removed a serving protocol ambiguity, but the final route still issued 20 actions serially, exhausted the turn cap, and failed at 98,543 tokens. GPT-4o mini also failed T28 at 107,896 tokens after focused-boundary repair. These admissions block the stronger marketing

Table 1: Preregistered eight-task, four-cell, n=10 substitution replication. Workflow 30210886840, commit 75a8c7911187. TPCA is tokens per correct answer and CoP is estimated API cost per passing task.

System	Correct	TPCA	CoP (USD)
gpt-5.6-terra raw	56/80	45,915	0.1642
gpt-5.6-terra + Specssmith FULL	80/80	11,737	0.0602
gpt-5.6-sol raw	65/80	28,023	0.1511
gpt-5.6-sol + Specssmith FULL	80/80	8,082	0.0752

claim that an 8B or mini model replaces a frontier model. They do not negate the separately tested lower-tier Terra system, whose parameter count and architecture are not inferred.

7 Threats to Validity

Construct validity. GovernanceBench measures repository-task execution under one controller, not general intelligence, developer productivity, code maintainability, or factual truth. Synthetic fixtures improve experimental control but may reward explicit contracts and validators more than naturally occurring repositories.

Internal validity. Proprietary model weights, serving stacks, and sampling implementations are not controlled. Repetition IDs do not guarantee identical provider randomness. Dollar costs are versioned estimates, not invoices. We mitigate these threats with same-commit matched cells, exact model-route labels, raw token receipts, fail-closed provider handling, and paired uncertainty.

External validity. Eight tasks are too few for an unconditional market-wide replacement claim. Task-cluster intervals expose this limitation but cannot manufacture diversity. The benchmark is public and may eventually be contaminated. Replication on fresh, real repositories such as SWE-bench-Live is necessary [14].

Researcher degrees of freedom. Earlier runs informed controller development, so the study includes adaptive optimization. The decisive release profile, thresholds, commit, and selected model pair were fixed before its run. We publish negative admissions and do not pool incompatible commits.

8 Operational Implications

The evidence motivates a guarded router rather than a single default model. Deterministic gates should handle tasks that do not require generation. Lower-tier governed models should receive bounded, validator-rich work for which they have passed admission. Frontier governed models remain appropriate for ambiguous localization, novel

architectures, and task families whose lower-tier confidence bound crosses the correctness or efficiency threshold. Routing is therefore based on verified marginal value, consistent with the cost-aware motivation of FrugalGPT and RouteLLM [2, 7], but grounded in executable software evidence.

9 Reproducibility and Ethics

The repository publishes the harness, fixtures, controller, exact workflow inputs, analysis code, compact per-cell measurements, and SHA-256 evidence manifest. Full traces are retained with the cited GitHub Actions runs according to the repository’s configured artifact-retention period; the compact evidence and source digests are versioned permanently. The paper reports model API identifiers but does not infer unpublished parameter counts or architecture. Hidden acceptance tests are isolated from model feedback. Safety tasks reward clarification or refusal rather than destructive execution. No human-subject data are collected.

10 Conclusion

Governance can move work from probabilistic model reasoning into deterministic requirements, retrieval, verification, and stopping. On the evaluated mixed task distribution, this substantially reduced the tokens required for correct software changes and allowed the governed Terra route to clear the preregistered substitution gates against raw Sol. The coding-only gate remained inconclusive by 0.7 percentage points, and the published-size admissions failed. The scientifically defensible result is task-conditional system substitution with explicit confidence and routing gates—not that smaller models universally replace frontier models.

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